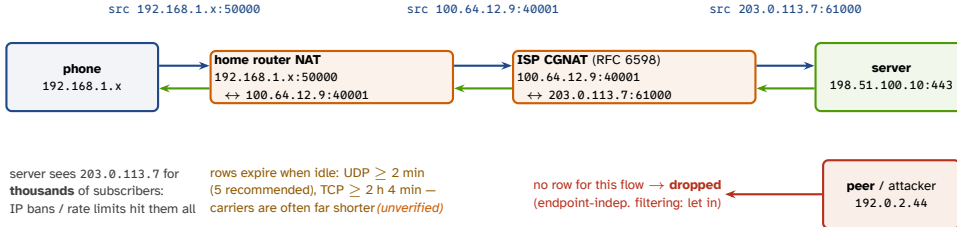


In one line: IPv4 ran out, so a phone sits behind **NAT** — often *two* (home router + carrier **CGNAT**) — which rewrites the private source **IP:port** to a shared public one and keeps a **mapping table**. Only *outbound* packets create rows, so nobody can dial *in*: P2P breaks, yet it is *not* a firewall. **IPv6** gives every device global addresses (no NAT); IPv6-only carriers reach IPv4 servers via **NAT64/DNS64**.

IPv4: home NAT + carrier CGNAT (NAPT)

blue: outbound creates the row and the source is rewritten at each hop (what the next hop sees, above) · **green:** the reply matches the row and is rewritten back



IPv6-only: DNS64 + NAT64

1 AAAA? api.example.com
2 DNS64: only A 198.51.100.10 → synthesise 64:ff9b::c633:640a
3 phone connects to that IPv6 address (it is fake)
4 NAT64 gateway strips the prefix → IPv4 198.51.100.10
c6.33.64.0a = 198.51.100.10
an IPv4 *literal* skips step 1 → no route. Use names.

IPv4 and NAT

- **32 bits** ≈ 4.3 bn addresses. IANA handed out its last /8s on 3 Feb 2011; APNIC hit its final /8 in Apr 2011, ARIN ran dry Sep 2015, RIPE NCC Nov 2019. NAT (really **NAPT**, port translation) lets a whole LAN share one address.
- **Not routed on the internet:** RFC 1918 10/8, 172.16/12, 192.168/16; **CGNAT** shared space 100.64/10 (RFC 6598); link-local 169.254/16. Docs/examples: 192.0.2/24, 198.51.100/24, 203.0.113/24 (RFC 5737).
- **Outbound packet** → NAT allocates public IP:port, stores the 5-tuple mapping, rewrites the source (and checksums). The **reply** matches the row and is rewritten back. No row → dropped. Idle rows **expire** (RFC 4787: UDP ≥ 2 min, 5 min recommended; RFC 5382: TCP established ≥ 2 h 4 min) — long-lived sockets need **keep-alives**.
- **CGNAT** on mobile: one public IP for many customers → per-IP rate limits and bans are collateral damage; abuse reports need IP + **port** + time; no port forwarding.

NAT behaviour — who may use a mapping

RFC 3489	RFC 4787: mapping · filtering	hole punch?
full cone	endpoint-indep. · endpoint-indep.	yes, easy
restricted	endpoint-indep. · address-dep.	yes
port-restr.	endpoint-indep. · address+port-dep.	yes (simultaneous)
symmetric	address(+port)-dep. · a+p-dep.	no → TURN

Mapping = does the public port stay the same for every destination? **Filtering** = who may send *in* on it. STUN only helps when the mapping is endpoint-independent: the port the STUN server saw is the port the peer can use. Symmetric NAT gives the peer a *different* port → relay. **Hairpinning**: two devices behind the same NAT reaching each other via the public address.

Why NAT is not a firewall — and asking it for a hole

Dropping unknown inbound is a *side effect* of having no row. Endpoint-independent filtering lets *any* host use an open mapping; nothing inspects outbound. IPv6 has no NAT, so the router needs a real **stateful firewall** (allow established, deny new inbound). **Port mapping**: static forwarding · **UPnP IGD** (unauthenticated) · **NAT-PMP** (RFC 6886) → **PCP** (RFC 6887) — none reaches through a carrier CGNAT.

Example — what an app must do

```
// BAD: IPv4 literal - no AAAA to synthesise on NAT64
let bad = URL(string: "http://198.51.100.10/v1")!
// GOOD: a name; DNS64 gives 64:ff9b::c633:640a
let good = URL(string: "https://api.example.com/v1")!
let mon = NWPathMonitor() // Network.framework
mon.pathUpdateHandler = { p in
    print(p.supportsIPv4, p.supportsIPv6) // IPv6-only: false, true
}
mon.start(queue: .main)
// test: macOS Internet Sharing -> "Create NAT64 Network"
// server: store IPs as 16 bytes; rate-limit per /64
```

IPv6

- **128 bits**; a LAN gets a /64: 64-bit prefix + 64-bit interface ID. No broadcast, no ARP (NDP/ICMPv6 instead — don't block ICMPv6).
- Every interface has **several** addresses: **link-local** fe80::/10 (always, never routed; needs a zone %en0) · **ULA** fc00::/7 (private, "RFC 1918 for v6") · **global** 2000::/3. Docs: 2001:db8::/32.
- **SLAAC** (RFC 4862): the router advertises the prefix (RA), the host builds its own address; DHCPv6 is optional. **Temporary (privacy) addresses** (RFC 8981) rotate the interface ID — outbound connections use them, so **an IP is not a device ID**.
- **Dual stack**: both families. **Happy Eyeballs v2** (RFC 8305): resolve A + AAAA, try IPv6 first, start the next attempt after 250 ms (50 ms resolution delay), keep the winner. URLSession / Network.framework race for you (*unverified*); a raw connect() to the first address does not.
- **IPv6-only + NAT64** (RFC 6146) with **DNS64** (RFC 6147) and the well-known prefix 64:ff9b::/96 (RFC 6052) — many mobile carriers. **464XLAT** (RFC 6877): an on-device CLAT gives IPv4-literal apps a fake IPv4 path. **Since 1 June 2016 every App Store app must work on IPv6-only networks**; Apple says: no IP literals, use getaddrinfo / URLSession.

Traps

- ! "NAT is security" — filtering is accidental; IPv6 security = the stateful firewall.
- ! "Two phones can just connect" — both behind NAT/CGNAT: neither can accept; STUN + hole punching, else **TURN** (symmetric NAT, UDP blocked).
- ! Idle TCP socket silently dead after minutes on cellular — the NAT row expired, no RST. Heartbeat below the timeout, or reconnect.
- ! Banning an IP on mobile = banning a neighbourhood (CGNAT).
- ! Hard-coded IPv4 literal / IPv4-only socket code = rejected by App Review on NAT64.

Remember

"NAT hides, it doesn't guard. Outbound opens the row, idle closes it. Names, never literals."